

Phylogeography of spiny mice (genus *Acomys*, Rodentia: Muridae) from the south-western margin of the Sahara with taxonomic implications

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The present study aims to depict the overall pattern of *Acomys* history in south-western Sahara. We tested the specific predictions that: (1) several mitochondrial clades can be identified and that they coincide with the described species; (2) successive phases of desert expansion and contraction during the last 3 Myr have resulted in several phases of demographic expansion and population fragmentation in *Acomys*; and (3) isolation-by-distance occurs. The extent of phylogeographic patterns and molecular genetic diversity (cytochrome *b* gene and D-loop) were addressed in a survey of 90 individuals of *Acomys* from 38 localities. Our phylogeographical analyses show a strong genetic structure within western Saharan *Acomys*, with several phylogroups displaying non-overlapping geographic distributions. Restricted gene flow with isolation-by-distance was recorded and a signal of population expansion was detected within several clades. We suggest that during arid or semi-arid paleoclimatic periods, when large sandy areas were present, *Acomys* was restricted to rocky massifs, whereas, in more humid periods, when savannah and/or steppe habitat prevailed, this species was able to disperse and to survive outside rocky areas because food resources were abundant. Based on a comprehensive sampling and the use of an integrative approach (i.e. combining cytogenetic, molecular and morphological data), we firmly propose that *Acomys airensis* should be considered as a junior synonym of *Acomys chudeaui*. © 2009 The Linnean Society of London, *Biological Journal of the Linnean Society*, 2009, 98, 29–46.

ADDITIONAL KEYWORDS: Africa – biogeography – cytogenetics – desert – morphology – pleistocene – rodent.

INTRODUCTION

The Sahara Desert is a major geographical feature of the African continent. It occupies slightly over 8 million km², between latitudes 16 and 32°N,

circumscribed within the 100 ± 50 mm isohyet of mean annual rainfall (Le Houérou, 1997). The timing of the onset of desert conditions in the Sahara is poorly known, although recent data suggest that it became hyper-arid approximately 7 Myr ago (Pickford, Wanas & Soliman, 2006; Schuster *et al.*, 2006; Carranza *et al.*, 2008). Since the Pliocene, however, the Sahara has experienced drastic climatic changes, rendering its boundaries highly variable. On the one

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hand, many regions that are now dry were once much more humid, sometimes harbouring steppe, savannah or even forest conditions. On the other hand, the desert has displayed periods of expansion, when the current Sahelian region was a wide desert band of sand dunes (Rognon, 1989). The frequency of the Saharan cycles of desert spread and retraction are estimated to have ranged from approximately 100 000–20 000 years during the last 3 Myr (Rognon, 1989). It is very likely that these relatively rapid and important paleoclimatic shifts have affected the geographic distribution and evolution of animal species.

Analysis of intraspecific phylogeographic patterns has led to major advances in our understanding of historical biogeographical processes (Avise, 2000). Published mitochondrial (mt)DNA studies of widespread organisms frequently describe levels of intraspecific divergence corresponding to Pleistocene or Pliocene population fragmentation/colonization events, which are interpreted to explain the currently observed patterns of genetic differentiation. Numerous studies were carried out in Europe and North America (Hewitt, 2004a, b; Schmitt, 2007), although few are available for Africa (Nicolas *et al.*, 2008a, b). Within Africa, phylogeographical studies concerning the Sahara region are scarce: one concerns the snake *Psammophis schokari* (Rato *et al.*, 2007) and another relates to the large mammal *Oryx dammah* (Iyengar *et al.*, 2007). Both studies identified several genetic lineages within each species that were hypothesized to have originated from expansion/contraction phases of the Sahara desert, associated with severe climate changes between the Miocene and Pleistocene. Moreover, a few studies have suggested that episodes of north–south vicariance speciation events were mediated by the Sahara temporal variations. The Sahara appears to have allowed dispersal from the tropical areas during humid phases of the Mio-Pleistocene but to have isolated populations north and south of it during more arid phases (Franck *et al.*, 2001; Douady *et al.*, 2003; Guillaumet, Crochet & Pons, 2008).

In general, small rodents are good candidates for inferring biotic history from contemporary patterns of genetic variation owing to their short generation time and rapid mtDNA substitution rate, their relatively limited dispersal ability, and their strong associations with particular habitats (Fedorov *et al.*, 2008). Moreover, their distributions are not affected by hunting and are less likely to have been modified by recent human activities.

In the present study, we investigated the phylogeography of a murid rodent of the genus *Acomys* in a wide area of the south-western fringe of the Sahara. In this region, the genus *Acomys* (spiny mice) comprises three cryptic species that are supposedly

distinguishable on dental traits: *Acomys airensis*, *Acomys chudeaui*, and *Acomys seuratii*. *Acomys airensis* is distributed in southern Mauritania, Mali, Niger, and Chad; *A. chudeaui* is known from central Morocco and western Mauritania; and *A. seuratii* is only recorded from mountains of southern Algeria, in Hoggar, Mouydir, and Tassali n'Ajers (Musser & Carleton, 2005). These species all belong to the *Acomys cahirinus–dimidiatus* species group (Musser & Carleton, 2005) and speciation events within this group are estimated to have occurred during the last 2.8 My (Barome, Monnerot & Gautun, 2000), thus making its differentiation clearly subsequent to the first onset of desert conditions in the Sahara. Most, if not all, *Acomys* species are mainly rock-dwellers, and are rarely found in other habitats (Volobouev *et al.*, 2002a). As such, discontinuities in their distribution may be expected, leading to genetic structuration of their populations.

The present study aimed to depict the overall pattern of *Acomys* history in south-western Sahara and to correlate past climate-driven landscape modifications with the genetic structure of these populations. The phylogenetic relationships among populations were determined by the analysis of cytochrome *b* (*cytb*) and D-loop sequence variation in numerous specimens collected in Mauritania, Mali, and Niger. We tested the specific predictions that: (1) several mitochondrial clades can be identified within the genus *Acomys* in south-western Sahara and that they coincide with the described species; (2) successive phases of desert expansion and contraction during the last 3 Myr have resulted in several phases of demographic expansion and population fragmentation (bottlenecks), respectively; and (3) isolation-by-distance occurs, particularly among populations inhabiting the main rocky areas.

MATERIAL AND METHODS

SPECIMENS EXAMINED AND MTDNA SEQUENCING

Tissue samples were obtained from *Acomys* from 38 localities (Fig. 1, Table 1). A total of 1093 bp of the *cytb* sequence were obtained for 90 individuals (one to eight individuals per locality) and a total of 563 bp of the D-loop were obtained for 83 individuals (one to eight individuals per locality).

DNA was extracted from ethanol-preserved tissues by the CTAB method (Winnepeninckx, Backeljau & De Wachter, 1993). The *cytb* gene was amplified using polymerase chain reaction (PCR) primers L14723 and 15915 (Ducroz, Volobouev & Granjon, 2001). The D-loop was amplified using PCR primers La1 5'-ATAAAAATTACTCTGGTCTTGTAAC-3' and Hb 5'-TGTCTTAATTTAGGGGAACG-3' (present study).

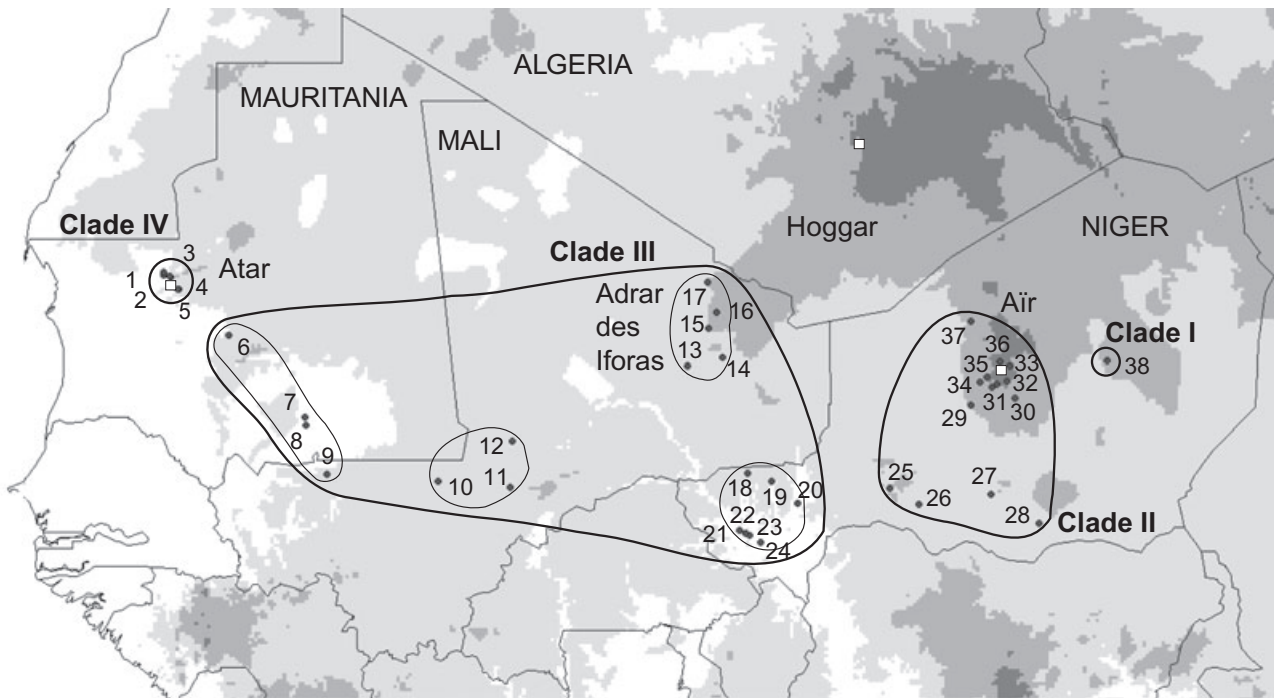


Figure 1. Representation of the four groups of populations (clades I–IV) recovered in the phylogenetic, network, and spatial analysis of molecular variance (SAMOVA) procedure analyses. The grey scale indicates the relief: white, < 250 m a.s.l., light grey, 250–500 m a.s.l., slightly dark grey, 500–1000 m a.s.l., dark grey, > 1000 m a.s.l.. Thin lines indicate the four sub-groups of populations recovered in the network and SAMOVA analyses. Black circles represent sampling sites (for sampling sites, see Table 1) and white squares indicate type localities of *Acomys chudeaui* (Mauritania), *Acomys airensis* (Niger), and *Acomys seurtati* (Algeria).

The PCR consisted of 35 cycles: 30 s at 94 °C, 40 s at 50–55 °C, and 90 s at 72 °C. The double-stranded PCR products were purified and sequenced at the Genoscope (Ivry Sur Seine, France). Sequences were aligned using CLUSTAL W (<http://align.genome.jp/>).

PHYLOGENETIC ANALYSIS

Evolutionary relationships among all haplotypes were estimated by constructing phylogenetic trees using maximum likelihood (ML) and Bayesian Markov-chain Monte Carlo phylogenetic analyses (MCMC). The ML analysis was performed with the PHYML software (Guindon & Gascuel, 2003) and Bayesian analysis with MrBayes, version 3.1 (Huelsenbeck & Ronquist, 2001).

The computer software MODELTEST, version 3.04 (Posada & Crandall, 1998) was used to evaluate the fit of 56 nested models of nucleotide substitution to the data. The model chosen by MODELTEST according to the Akaike information criterion was then used in ML and Bayesian analyses. MODELTEST revealed that the model which best fitted the data for the D-loop was the TrN + I + G. However, because this model is not implemented in the MrBayes software,

we used the GTR + I + G model of substitution in the MCMC analysis. Both the TrN and GTR models consider separate frequency for each base. However, although the TrN model considers only three distinct substitution rate, the GTR model considers six distinct substitution rates. ML analyses were performed on the PHYML online web server (Guindon *et al.*, 2005). In the PHYML procedure, we used as starting tree the BIONJ distance-based tree. Bootstrap analysis (500 replicates) was used to estimate the robustness of internal nodes. A clade was considered as well supported when the bootstrap support was higher than 80%.

Bayesian analyses were conducted *sensu* Nicolas *et al.* (2008a). A clade was considered as well supported when the posterior probability was higher than 0.95.

Phylogenetic analyses were conducted for each gene separately and for both genes combined. Our phylogenetic trees were first rooted with as many *Acomys* species as possible (i.e. *Acomys subspinosus*, *Acomys spinosissimus*, *Acomys russatus*, *Acomys johannis*, *Acomys dimidiatus*, *Acomys cilicius*, *Acomys cahirinus*, and *Acomys minous*; one specimen par species; for GenBank numbers, see Supporting information,

Table 1. List of the sampled localities, with the number of specimens sequenced, the karyotype (2n and FNa), and the haplotypes codes for each mitochondrial region in each locality

Country	Locality	Locality number	2n	FNa	Clade	Cytochrome <i>b</i>		D-loop	
						Sample size	Haplotypes	Sample size	Haplotypes
Niger	Fachi	38	46	66	I	2	h50, h51	2	h47
Niger	Agadez, Alacess	29	42	66	II	1	h09	1	h01
Niger	Agadez, Indoudou	30	42	66	II	1	h16	1	h06
Niger	Aouderas	35	43	66	II	3	h01, h12, h16	3	h02, h07, h09
Niger	Gangara	27	46	66	II	1	h02	1	h04
Niger	Guimat	34			II	2	h14, h15	2	h07, h09
Niger	Hamdara	28	43	66	II	2	h05	1	h16
Niger	Idecle	31			II	1	h13	1	h06
Niger	Keita	25			II	3	h08, h10	2	h05, h13
Niger	Mt Baguezan	32			II	2	h14, h15	2	h10
Niger	Ourou	37	42	66	II	1	h17	1	h12
Niger	Sodané	26			II	2	h03, h04	2	h03
Niger	Tabelot	33	40, 42	66	II	3	h12, h14, h18	3	h07, h11, h14
Niger	Timia	36			II	5	h06, h07, h11, h14	5	h07, h08, h15
Mali	Amazan	11			III	1	h41	0	
Mali	Dazinga	24			III	1	h20	0	
Mali	Edjerir	14			III	1	h38	1	h27
Mali	Farabougou	10	40	66	III	3	h43	3	h36
Mali	Gakou	9	40	66	III	4	h29	4	h33
Mali	Niodougou	12	40	66	III	1	h42	1	h36
Mali	Terarabat	16	44, 46	66	III	2	h37, h39	2	h24, h26
Mali	Tessalit	17	46	66	III	3	h40	3	h25
Mali	Tilemsi	13			III	1	h40	0	
Mali	Timajelalin	15			III	1	h39	1	h24
Mauritania	Oumcreya	7			III	6	h30, h31	6	h32, h34, h35
Mauritania	Chalkha	8			III	1	h32	0	
Mauritania	Rachid	6			III	8	h28, h34, h35, h36	8	h28, h29, h30, h31
Niger	Batchintoulou	20	42	66	III	1	h19	1	h18
Niger	Karma	21	41	66	III	2	h26, h27	1	h20
Niger	Niamey	23			III	2	h21, h26	2	h17, h20
Niger	Sumett	19	42	66	III	1	h23	1	h22
Niger	Tilola	18	40, 42	66	III	3	h22, h24	3	h21, h23
Niger	Tondibia	22	42	66	III	2	h25	2	h19
Mauritania	Aoudjeft	5			IV	1	h49	1	h38
Mauritania	Ayn Ehel Taya	2	40	66	IV	2	h47	2	h40
Mauritania	Hamdoun	1			IV	4	h44, h46, h48	4	h39, h43, h45, h46
Mauritania	Souegya	4			IV	4	h45, h46	4	h37, h41
Mauritania	Tergit	3	40	66	IV	6	h44, h45	6	h42, h44

Table S1), and by two other genus members of the Deomyinae subfamily (*Deomys* and *Uranomys*) (Dubois, Catzeflis & Beintema, 1999; Stepan *et al.*, 2005). These preliminary analyses revealed that *A. minous*, *A. cilicius*, *A. cahirinus*, *A. johannis*, and *A. dimidiatus* were the sister species group of our species of interest. Thus, only these species were retained as outgroups in our final analyses. Phylogenetic relationships between haplotypes were also inferred by constructing a network using the median-joining method available in NETWORK, version 4.500 (Bandelt, Forster & Rohl, 1999).

We examined the *cytb* and D-loop data set for saturation. Using the matrices of patristic and inferred substitutions calculated by PAUP 4b10 (Swofford, 2000), the pairwise numbers of observed differences was plotted against the corresponding values for inferred substitutions using the software TransPAUPMatrix (<http://lis.snv.jussieu.fr/apps/contrib/TPM/TransPAUPMatrix.htm>). Because of differential rates at codon positions, saturation plots for *cytb* were analysed separately at the first, second, and third codon position.

ESTIMATES OF GENETIC DIVERSITY, GENETIC STRUCTURE, AND HISTORICAL DEMOGRAPHY

All these analyses were conducted for each gene separately. We applied the spatial analysis of molecular variance procedure (SAMOVA; Dupanloup, Schneider & Excoffier, 2002) to identify groups of geographically adjacent populations that were maximally differentiated based on sequence data. Using the computer software SAMOVA, version 1.0, we performed analyses based on 100 simulated annealing steps and examined maximum indicators of differentiation (F_{CT} values) when the software was instructed to identify $K = 2$ to $K = 12$ partitions of populations. We analysed population structure by analysis of molecular variance (AMOVA) in ARLEQUIN, version 3.11 (Excoffier, Laval & Schneider, 2005). AMOVA divides the total variance into additive components (i.e. variation within populations, among populations within groups and among groups). A population was defined as all individuals coming from one geographical locality; and groups of populations were defined according to the results of the SAMOVA analysis.

Where applicable, we estimated several statistics to describe and compare the major clades recovered from our phylogenetic analyses. Nucleotide diversity and haplotype diversity were calculated for each clade using DNASP, version 4.50.2 (Rozas *et al.*, 2003).

Population history was assessed using an array of statistics that were introduced as tests for neutrality but are also capable of detecting the genetic impacts of population growth, decline or stability: Tajima's D

(Tajima, 1989), Fu and Li's F^* and D^* (Fu & Li, 1993) and Fu's F_s test (Fu, 1997). These analyses were performed with ARLEQUIN, version 3.11 (Excoffier *et al.*, 2005) or DNASP, version 4.50.2 (Rozas *et al.*, 2003). As suggested in the ARLEQUIN manual, the F_s statistics were considered significant when $P < 0.02$. The significance of the neutrality statistics was tested by generating random samples under the hypothesis of selective neutrality and population equilibrium, using a coalescent simulation algorithm adapted from Hudson (1990).

To evaluate the hypothesis of recent population growth, mismatch distributions were also calculated (Rogers & Harpending, 1992) as the pairwise differences between all haplotypes within each clade using ARLEQUIN, version 3.11 (Excoffier *et al.*, 2005). The fit of our dataset to the expectations of the demographic expansion model (sudden expansion model) or the spatial expansion model (infinite-island model) was tested using the sum-of-squared deviations between the observed and expected mismatch from 1000 parametric bootstrap replicates.

The plausibility of an isolation-by-distance scenario was explicitly tested by performing Mantel's tests (Mantel, 1967) using ARLEQUIN, version 3.11 (Excoffier *et al.*, 2005) to analyse the relationships between genetic (mean number of pairwise nucleotide differences) and geographical distances between populations. The geographical distances between localities were calculated from GPS coordinates in the software GENALEX, version 6 (Peakall & Smouse, 2006). Data were permuted 1000 times to estimate the 95% upper tail probability of the matrix correlation coefficients.

TIMES OF DIVERGENCE

Using the clades identified in the phylogenetic analyses, a Bayesian MCMC approach was used to calculate the time to the most recent common ancestor (TMRCA). TMRCA estimates when genes last shared a common ancestor and therefore can be used as a proxy for ancestral population age. Clades were dated using the software BEAST, version 1.4.6 (Drummond & Rambaut, 2007) *sensu* Nicolas *et al.* (2008a). MODELTEST was used to identify the model of sequence evolution used in BEAST, version 1.4.6, and a Bayesian skyline coalescent model of population size was specified (Drummond *et al.*, 2005). Because *Acomys* has a poor fossil record, calibration of a molecular clock for this genus is not possible; instead, the *Mus/Rattus* lineage split, based on fossil records and estimated as 11–12.3 Myr ago, was employed (Benton & Donoghue, 2007). Two sequences of *Mus* and *Rattus* extracted from GenBank were also included in our analyses (*Mus musculus* J01420, *Rattus norvegicus*

141848). For the estimation of the TMRCA of the main clades recovered in the phylogenetic analyses, only the *cytb* sequence data were used. For the D-loop sequences, the alignment of outgroups was ambiguous and required the addition of numerous indels.

RESULTS

HAPLOTYPE VARIATION

Among the 90 ingroup *cytb* sequences, 51 different haplotypes were identified: 133 sites (12%) were variable and 76 (7%) were parsimony informative.

Among the 83 ingroup D-loop sequences, 47 different haplotypes were identified: 64 sites (11%) were variable and 32 (6%) were parsimony informative. The alignment of the 83 ingroup sequences required one indel.

No evidence of saturation for transitions or for transversions was detected within our *cytb* and D-loop data sets (see Supporting information, Fig. S2).

PHYLOGENETIC ANALYSIS

For both the *cytb* gene and the D-loop, trees obtained by ML and Bayesian analyses had a similar topology; consequently, only trees obtained by ML are shown in Figure 2.

For the *cytb* gene, results of the phylogenetic and network analyses are congruent (Figs 2, 3) and highlight the existence of several clades within our sample, with a strong geographical structure: specimens from one clade always come from adjacent localities on the map (Fig. 1). Four main geographical clades can be identified:

- one well-supported clade (clade I) in the locality of Fachi (Niger, Locality 38)
- one well-supported clade (clade II) in Niger (localities 25–37)
- one moderately-supported clade (clade III) in eastern Mauritania (localities 6–8), Mali (localities 9–24), and western Niger (localities 18–24)
- one well-supported clade (clade IV) in western Mauritania (localities 1–5).

Clades II and III form a highly supported monophyletic group (Fig. 2), and this clade is the sister clade of clade IV.

For the D-loop, the same four groups of haplotypes as for the *cytb* gene are identified in the network analysis (Fig. 1). However, only one of them (clade II) is well supported in the phylogenetical analyses. The phylogenetic trees suggest that clade II is derived from within clade III. As with the *cytb* gene, haplotypes corresponding to clades II, III, and IV form a highly supported monophyletic group.

For the combined data set (*cytb* + D-loop), trees obtained by ML and Bayesian analyses have the same topology than those obtained for the *cytb* data set (see Supporting information, Fig. S3).

NUCLEOTIDE DIVERSITY AND GENETIC STRUCTURE

For the *cytb* gene, the mean percentage of sequence divergence (K2P) between clades ranges from 1.45 (between clades II and III) to 4.08 (between clades I and II; Table 2). Within clades, nucleotide diversity $\times 100$ ranges from 0.188 ± 0.031 to 0.673 ± 0.075 and haplotype diversity ranges from 0.787 ± 0.075 to 0.968 ± 0.011 (Table 3).

For the D-loop, the mean percentage of sequence divergence between clades ranges from 1.87 (between clades III and IV) to 5.26 (between clades I and II; Table 2). Within clades, nucleotide diversity $\times 100$ ranges from 0.495 ± 0.066 to 0.965 ± 0.043 and haplotype diversity ranges from 0.831 ± 0.085 to 0.960 ± 0.023 (Table 3).

The results from the SAMOVA analyses, as expected (Dupanloup *et al.*, 2002), showed that when the number of groups of populations (K) increased, the value of F_{CT} increased (Fig. 4). F_{CT} only increases in small increments for $K = 4$ to $K = 12$, suggesting that adding extra groups only moderately improved the model of population structure. The group membership at $K = 4$ exactly matched those found in the phylogenetic and network analyses (clades I, II, III, and IV; Fig. 3) and was the same for both mtDNA regions. Genetic structure was highly structured among these groups of populations (AMOVA; permutation tests highly significant, $P < 0.00001$), meaning that most of the genetic variability was partitioned among groups (71.56% and 63.66% for the *cytb* and D-loop mtDNA regions, respectively) and only a small proportion of the variability was partitioned among populations within groups (18.91% and 20.00%) or within populations (9.53% and 16.33%).

We tested for signatures of population expansion in clades II, III, and IV. For both mtDNA regions, the results obtained through the neutrality tests and the mismatch distributions were congruent and did not

Table 2. Mean genetic distance (K2P) between clades. Below diagonal: results obtained for the D-loop; above diagonal: results obtained for the cytochrome *b* gene

	I	II	III	IV
I	–	4.08	3.91	3.70
II	5.26	–	1.45	2.67
III	4.42	2.32	–	2.60
IV	3.81	2.38	1.87	–

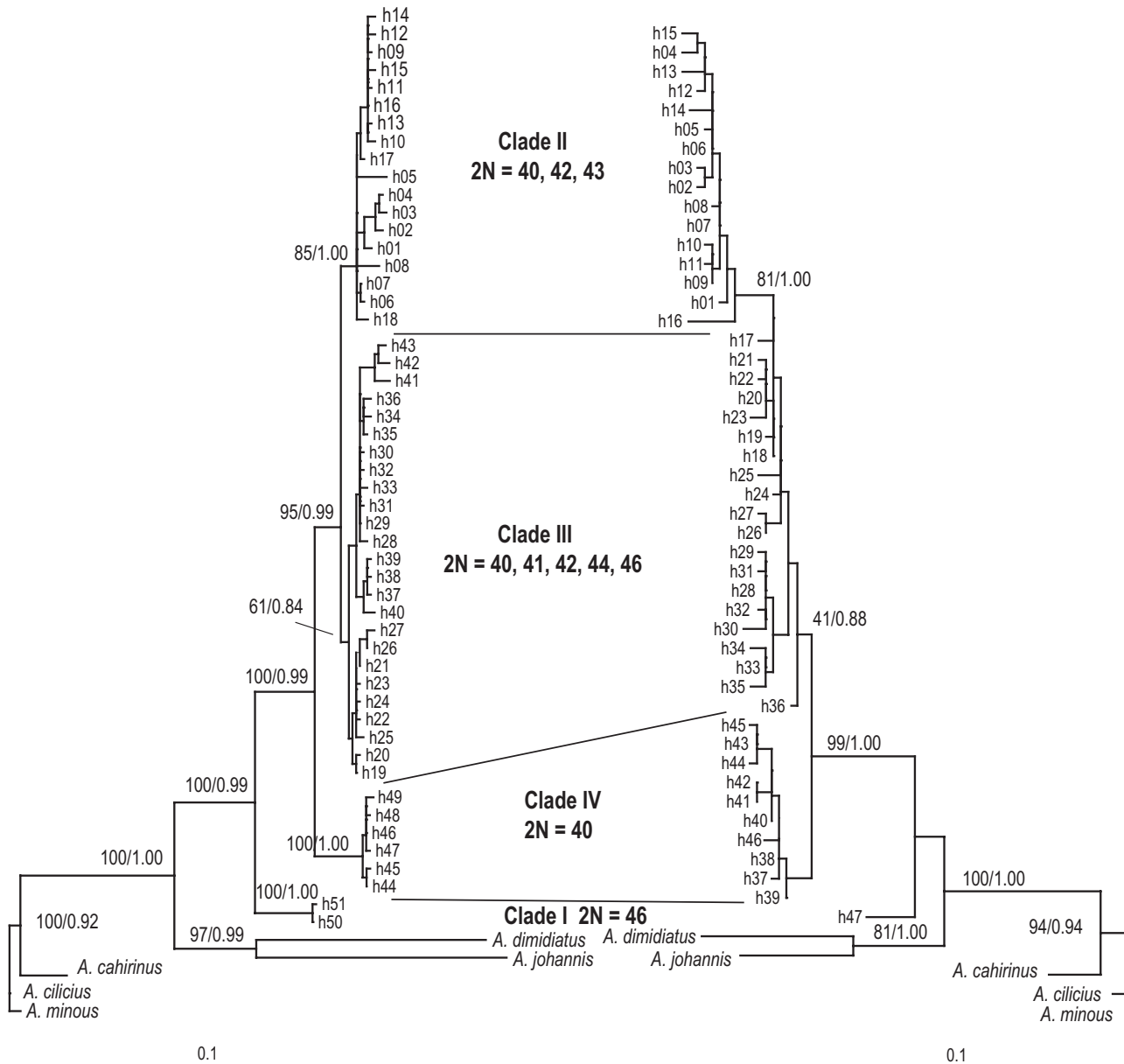


Figure 2. Phylogeny recovered by the maximum likelihood (ML) analysis for the cytochrome *b* gene (left, GTR + G model) and the D-loop (right, TrN + I + G). Numbers above nodes represent ML bootstrap support (500 replications) and Bayesian posterior probabilities respectively. To improve clarity, posterior probabilities for the most apical nodes are not shown. Locality information and haplotype designations are given in Table 1. Chromosomal data ($2n$) are derived from bibliographical data, as well as from our own data.

indicate any signal of population expansion in clade IV. For both mtDNA regions, the results obtained through the Fu's F_s test and the mismatch distributions were congruent and indicated a clear signal of population expansion in clade III. Even if it did not differ significantly from zero, the negative value of Tajima's D indicated a possible population expansion in clade III. For this clade, the pure demographic expansion model and the range expansion model have

virtually identical expected mismatch distributions (Fig. 5).

For clade II, a signal of population expansion was obtained with the D-loop sequence data, but not with the *cytb* sequence data. For this clade, the pure demographic expansion model and the range expansion model have virtually identical expected mismatch distributions with respect to the D-loop sequence data. For the *cytb* sequence data a multimodal mismatch

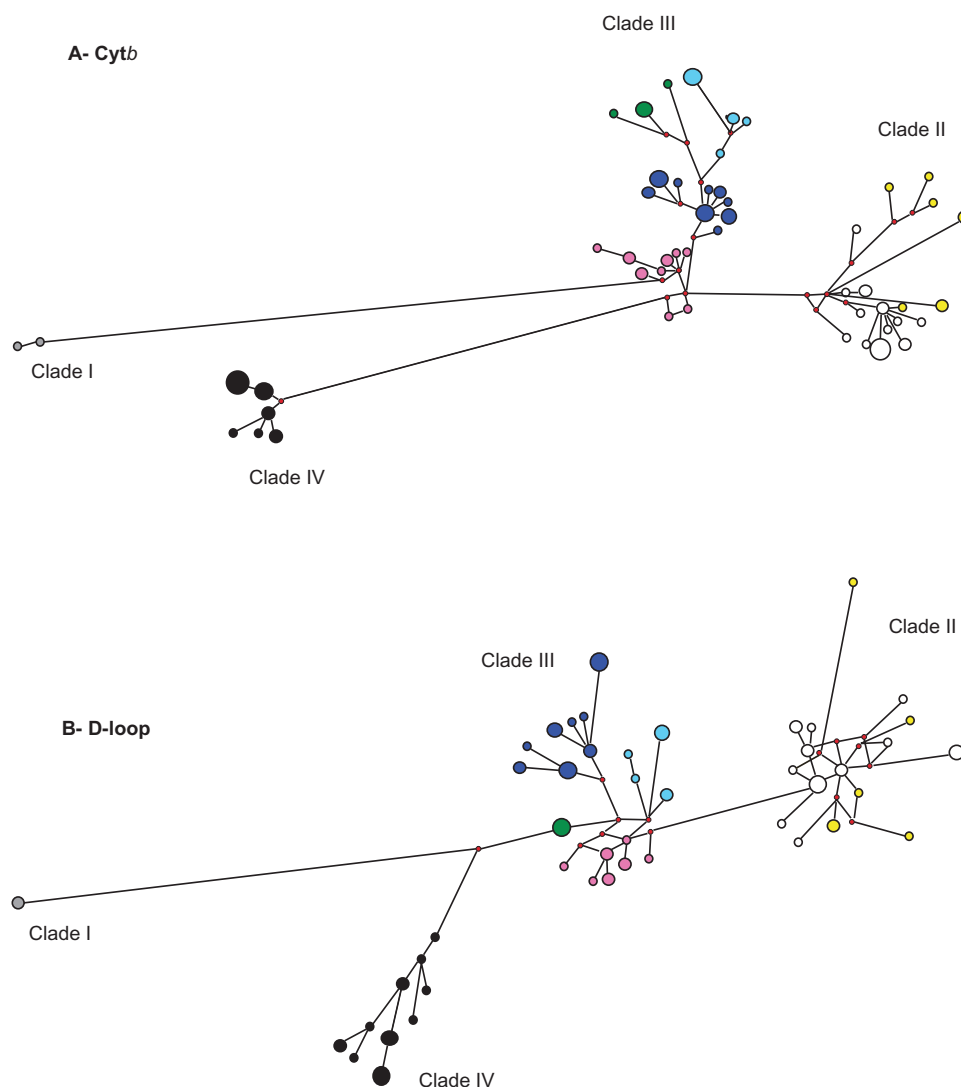


Figure 3. Median-joining network obtained for the cytochrome *b* gene (A) and the D-loop (B). Circle sizes are proportional to the number of the same haplotypes observed in the data set. Branch lengths are proportional to the number of mutations between haplotypes. For haplotype designations, See Table 1. Colours indicate geographical origin of the haplotypes: white, Air; yellow, south central Niger; pink, south west Niger; light blue, Adrar des Iforas; dark blue, southeastern Mauritania; green, south central Mali; black, Adrar (Mauritania); grey, Fachi. Small red circles indicate missing or unsampled haplotypes.

distribution was observed and neither the pure demographic expansion model, nor the range expansion model fitted well to the observed distribution (Fig. 5).

For both mtDNA regions, the network analysis did not display a clear starlike pattern in clades II and III, and numerous ancestral haplotypes were missing or unsampled (red dots in Fig. 3). Moreover, within clade III, some spatial clustering of haplotypes was observed (Figs 1, 3). These results suggest that it would be necessary to increase sample size in each locality to better investigate the observed pattern of demographic and/or spatial expansion and to identify the possible centres of expansion.

For both mitochondrial regions, a significant relationship was found between geographical and genetic distances (Mantel test, $P < 0.0001$, $r = 0.510$ and 0.494 for the *cytb* gene and the D-loop, respectively; Fig. 6). No significant relationship between geographical and genetic distances was found in clades II and III (Mantel test, $P > 0.05$; Fig 5). However, in the network analysis, some spatial clustering was observed within clade III (Figs 1, 3).

DIVERGENCE TIME ESTIMATES

The TMRCA of all south-western Saharan *Acomys* sequences was estimated at 1.252 (range

Table 3. Diversity and neutrality estimates for the main clades identified in the phylogenetic analyses

	All clades	Clade I	Clade II	Clade III	Clade IV
<i>Cytochrome b</i>					
<i>N</i>	90	2	27	44	17
<i>N_P</i>	133	1	45	48	7
<i>N_H</i>	51	2	18	25	6
<i>H_D</i>	0.982 ± 0.005		0.957 ± 0.025	0.968 ± 0.011	0.787 ± 0.075
<i>Pi</i> × 100	1.584 ± 0.099		0.673 ± 0.075	0.658 ± 0.048	0.188 ± 0.031
<i>k</i>	17.308		7.305	7.188	2.059
Fu & Li's <i>D</i> *			−0.864	−0.976	−0.477
Fu & Li's <i>F</i> *			−1.218	−1.265	−0.404
Fu's <i>F_s</i>			−4.904	−8.006	−0.491
Tajima's <i>D</i>			−1.399	−1.221	−0.019
<i>D-loop</i>					
<i>N</i>	83	2	25	39	17
<i>N_P</i>	64	0	20	29	11
<i>N_H</i>	47	1	16	20	10
<i>H_D</i>	0.980 ± 0.005		0.957 ± 0.023	0.960 ± 0.013	0.831 ± 0.085
<i>Pi</i> × 100	1.803 ± 0.100		0.725 ± 0.094	0.965 ± 0.043	0.495 ± 0.066
<i>k</i>	10.135		4.073	5.430	2.779
Fu & Li's <i>D</i> *			−0.140	−0.242	−0.835
Fu & Li's <i>F</i> *			−0.414	−0.478	−0.765
Fu's <i>F_s</i>			−7.277	−6.115	−2.589
Tajima's <i>D</i>			−0.834	−0.718	−0.221

Number of specimens (*N*), number of polymorphic sites (*N_P*), number of distinct haplotypes (*N_H*), haplotype diversity (*H_D*), nucleotide diversity (*Pi*), average number of nucleotide differences (*k*). Nucleotide diversities are expressed as percentages (i.e. 0.001 = 0.1%). Significant values are shown in bold.

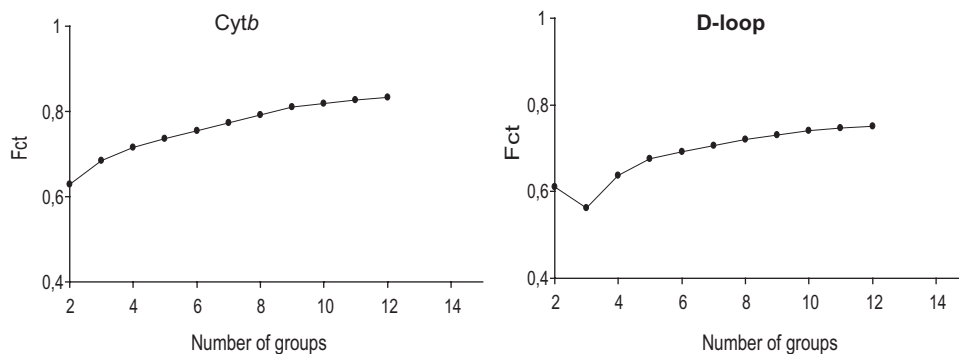


Figure 4. Plot of the F_{CT} parameter for different values of *K*, the number of population groups, generated using spatial analysis of molecular variance (Dupanloup *et al.*, 2002).

0.654–1.943) Mya. The TMRCA for clades II and III were estimated at approximately 0.250 (TMRCA = 0.244, range 0.118–0.395) and 0.283 (range 0.139–0.454) Mya, respectively. The TMRCA of clade IV was found to be lower [TMRCA = 0.121 (range 0.041–0.212) Mya] and was even more recent for clade I [TMRCA = 0.062 (range 0.005–0.118) Mya]. The TMRCA was estimated at 0.411 (range 0.196–0.642) Mya for clades II + III and at 0.704 (range 0.360–1.110) Mya for clades II–IV.

DISCUSSION TAXONOMY

Acomys chudeaui was described (Kollman, 1911) from Atar in Mauritania (20°31'N, 13°03'W), and is cited from central Morocco and western Mauritania. Then *A. airensis* was described (Thomas & Hinton, 1921) from Asben (Air, Mt Baguezan, Niger, 18°N, 8°30'E), and is cited from southern Mauritania, Mali, Niger, and Chad (Musser & Carleton, 2005).

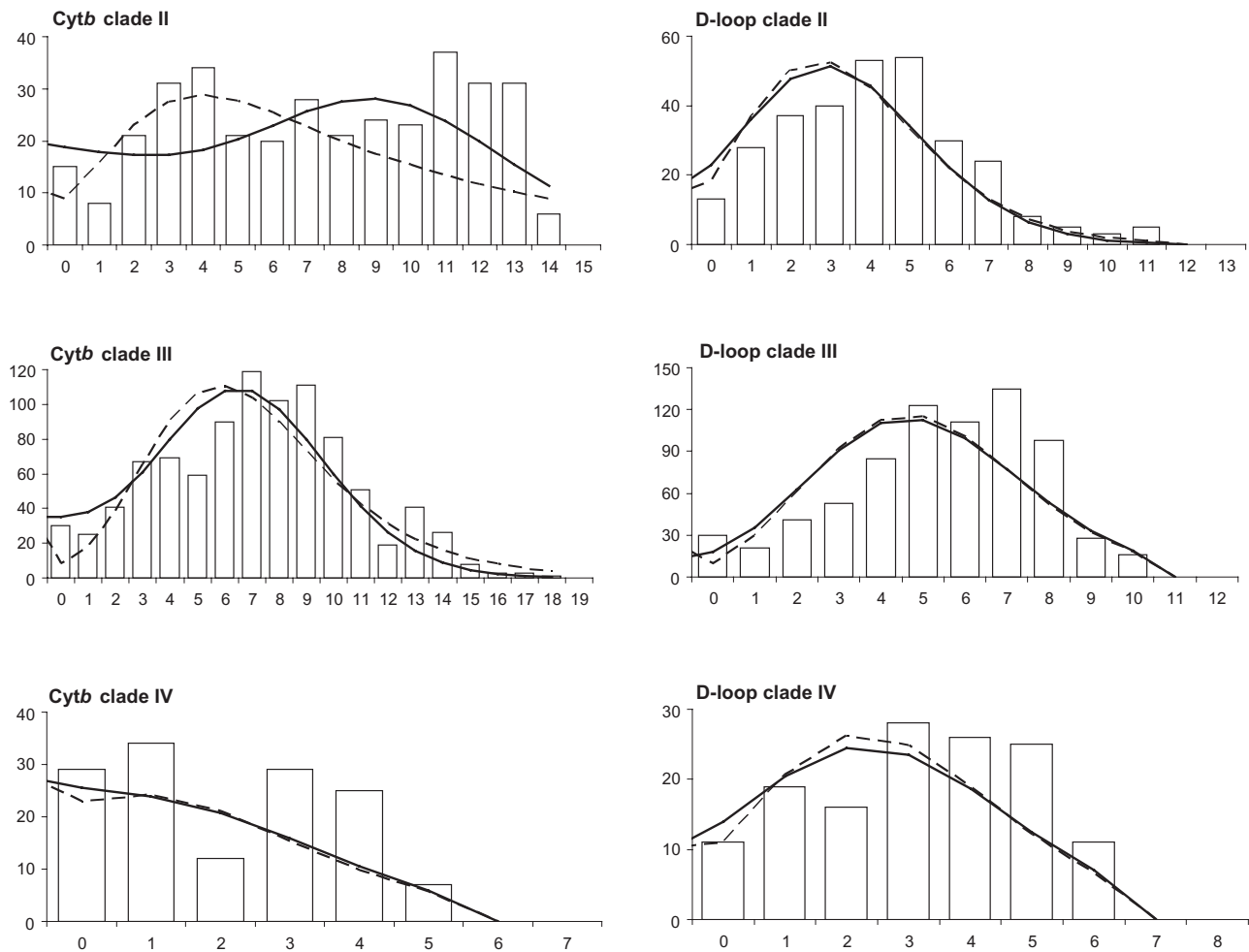


Figure 5. Observed and expected mismatch distributions under different models. White bars, observed distributions; solid lines, pure demographic expansion in an unsubdivided population model; dashed lines, range expansion in the infinite-island model.

Later, Heim de Balsac (1936) described *A. seurati* from mountains of southern Algeria and its karyotype was further established by Matthey & Baccard (1967) as $2n = 38$, $FN = 68$, and its molar morphology studied (Denys *et al.*, 1994). Unfortunately, we were unable to obtain samples from this region, and test the validity of this species from a molecular point of view. Conversely, the geographical origin of specimens of our clades II and IV makes them likely to represent *A. airensis* (Mt Baguezan, Air Massif in Niger) and *A. chudeaui* (Hamdoun, Tergit, Ayn Ehel Taya, and Souegya are less than 30 km south of Atar, Adrar Massif, and Mauritania), respectively (Fig. 1).

Descriptions of these two species were based on a few individuals and, in its description *A. airensis*, was not compared with *A. chudeaui*. As noted by Musser & Carleton (2005: 1195), 'The precise phylogenetic relationships among [these species] in which $2n$

varies from 40 to 46 requires resolution'. Indeed, karyotypic comparisons have been extremely useful for addressing taxonomic issues in various groups of African rodents (Duplantier, Britton-Davidian & Granjon, 1990; Dobigny, Aniskin & Volobouev, 2002a; Granjon *et al.*, 2002; Volobouev *et al.*, 2002a, b; Dobigny *et al.*, 2003).

Acomys airensis belongs to the so-called 'dimidiatus-cahirinus' complex of cryptic species (Barome *et al.*, 2000) that has evolved essentially through centric (or Robertsonian) fusions (Volobouev, Gautun & Tranier, 1996; Volobouev *et al.*, 2002a, 2007). Furthermore, it has previously been suggested that *A. airensis* may display such variations at the infraspecific level. Indeed, although its autosomal fundamental number remains constant ($FN_a = 66$), its diploid number was shown to range from $2n = 40$ to 46: $2n = 40$, 42, and 43 in the Air Massif (Tranier, 1975; Viegas-Péquignot *et al.* 1983;

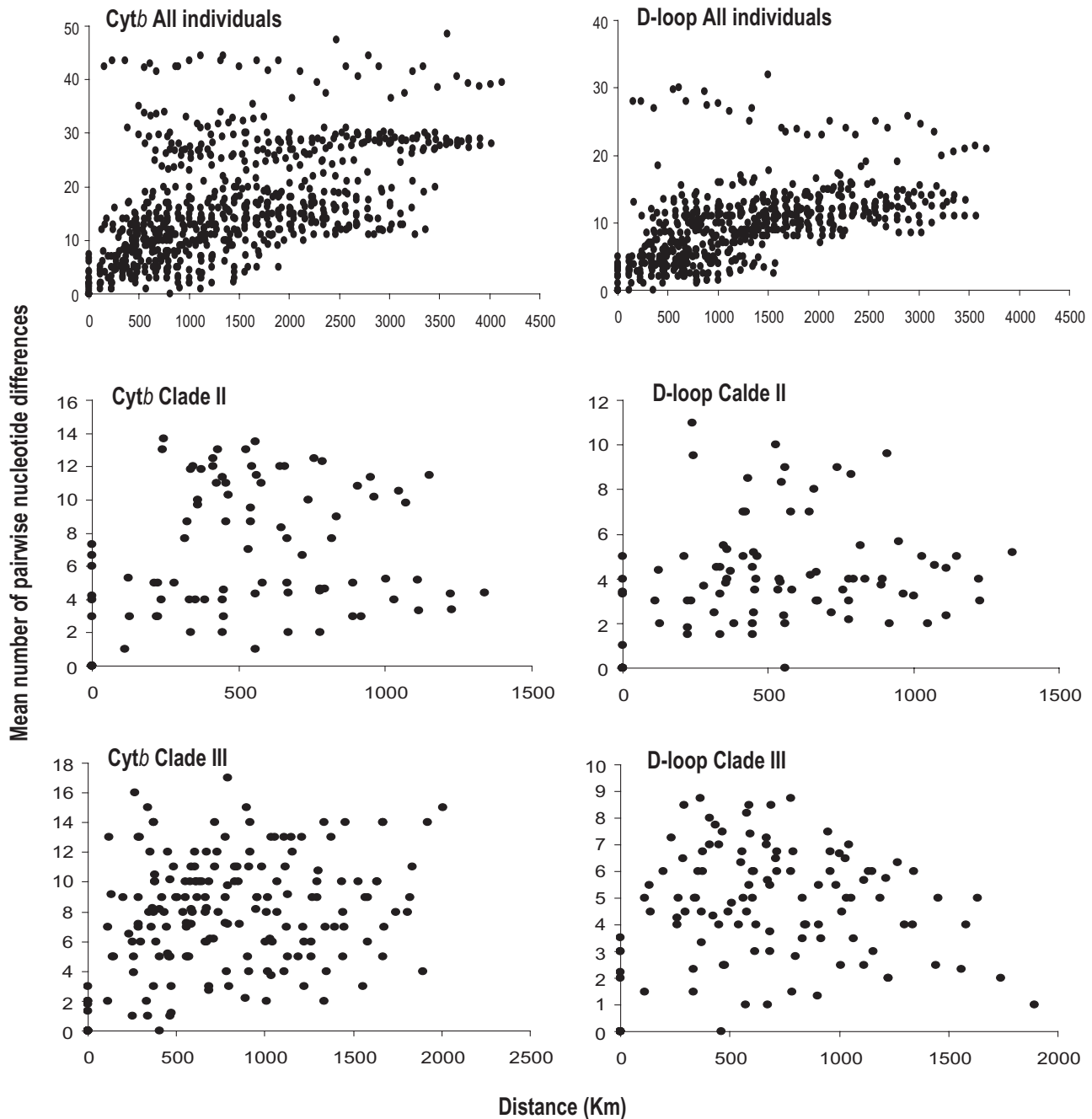


Figure 6. Scatter plots of geographical distances (km) and genetic distances (mean number of nucleotide differences).

Dobigny, Nomao & Gautun, 2002b; G. Dobigny, unpubl. data), $2n = 40, 41, 42, 43$, and 46 in the Sahelian area of Niger (Dobigny *et al.*, 2002b; G. Dobigny, unpubl. data, V. Volobouev, unpubl. data), $2n = 40$ in Sahelian parts of Mali (Tranier *et al.*, 1999; L. Granjon & B. Sicard, unpubl. data), $2n = 44$ and 46 in Adrar des Iforas (Dobigny *et al.*, 2001a, b), $2n = 40$ in south-eastern Mauritania (Bâ *et al.*, 2001), and $2n = 46$ in Fachi (Dobigny *et al.*, 2002b).

Interestingly, most of the latter individuals were also included in the present molecular analysis (Fig. 2, Table 1; see also Supporting information, Table S1), thus allowing for an unambiguous comparison of karyotypic and DNA-based data. Doing so, we are able to clearly demonstrate that these $2n$ variations are not clade-dependent, and thus correspond to intraspecific (most probably Robertsonian) variations. However, and critical to the present

study, the X and Y chromosomes were both found to be acrocentrics in all instances.

The first mention of *A. airensis* karyotype dates back to Tranier's (1975): $2n = 42$ chromosomes in specimens from Agades (south of the Air Massif, Niger). Subsequently, although before the description of the $2n = 40\text{--}46$ variation in *A. airensis*, one specimen from Tata (Morocco, $29^{\circ}44'N$; $7^{\circ}56'W$) and referred to as *A. chudeaui* (Benazzou, 1983) presented $2n = 40$ chromosomes, including 14 pairs of meta- or submetacentric and five pairs of acrocentric autosomal elements (FNa = 66). The X and Y were, respectively, identified as a submetacentric and a very small acrocentric chromosomes (FN = 69) (Benazzou, 1983). Taking into account the variation described in *A. airensis*, it now appears that this $2n = 40/\text{FNa} = 66$ karyotype most probably displays an 'airensis-like' autosomal complement. The only difference would then consist in the morphology of the X chromosome (submetacentric in *A. chudeaui* versus acrocentric in *A. airensis*).

Unfortunately, no published photograph of this karyotype, nor any biological material related to this specimen, is available, and we can only speculate about it. It may be the result of either: (1) a structural change, such as a centromeric shift, a pericentric inversion, and/or an addition/deletion of heterochromatin, or (2) an inaccurate interpretation of the conventionally stained karyotype of *A. chudeaui*.

Centromeric shifts, pericentric inversions, and the addition/deletion of heterochromatin have sometimes been described as species-specific characters (Trifonov *et al.*, 2008), proposed to induce reproductive isolation (Thelma *et al.*, 1988) or to favour evolutionary divergence (Volobouev *et al.*, 2002a) in various mammalian taxa. At the same time, polymorphic changes in X morphology have also been described in different lineages of the same species (Rao *et al.*, 1983; Paresque *et al.*, 2007). Consequently, if a true difference in X morphology exists between the so-called *A. airensis* and *A. chudeaui*, it would anyway be premature to conclude about its biological significance. However, in the absence of banding data, the NF difference more plausibly corresponds to a chromosome of the complement, something that would perfectly fit to both the *A. airensis* X morphology, and the $2n$ variation range that has been described subsequent to the study by Benazzou (1983). Supporting our view, it is noteworthy that all X chromosomes described to date in the 'dimidiatus-cahirinus' complex of species are acrocentric (Denys *et al.*, 1994: table 4; Granjon L., Dobigny G. unpubl. data). Moreover, two karyotyped individuals from Tergit and Ayn Ehel Taya [i.e. in the very close vicinity of *A. chudeaui* type locality (Atar, Mauritania)], both displayed the $2n = 40/\text{FNa} = 66$ karyotype and an acrocentric X chromosome that characterizes *A. airensis*.

From there, we suggest that Mauritanian spiny mice all possess a $2n = 40\text{--}46/\text{FNa} = 66$ karyotype that corresponds to *A. airensis*. Hence, this strongly suggests that *A. chudeaui* is also characterized by this karyotype. As a corollary, the $2n = 40$ karyotype described by Benazzou (1983) would then either correspond to a misinterpretation of the same complement (i.e. *A. airensis* with $2n = 40$), or, alternatively, belong to an undescribed species that is different from both *A. airensis* and *A. chudeaui*.

The percentage of sequence divergence (K2P) between clades II (*A. airensis*) and IV (*A. chudeaui*) is low (2.67% for the *cytb* gene) and similar to those observed within rodent species and lower than those usually observed between species. For example, the K2P percentage of sequence divergence between clades of *Mastomys huberti* can reach 3.08% (Mouline *et al.*, 2008), and varies from 1–5% within several species of south American rodents (Costa, 2003). The interspecific *cytb* sequence variation detected in *Peromyscus* (Harris, Rogers & Sullivan, 2000) and *Sigmodon* (Peppers & Bradley, 2000) rodents approaches 10%, varies between 7% and 18% within genera of the *Praomys s.l.* group (Lecompte *et al.*, 2002), and varies from 7.4% to 9.6% between species of the so-called *Acomys cahirinus-dimidatus* group (Barome *et al.*, 1998). Using mtDNA genetic distances to assign taxonomic rank (Johns & Avise, 1998; Bradley & Baker, 2001) is problematic, and is not without its detractors (Ferguson, 2002). However, these genetic distance data when combined with the karyotypic information detailed above suggest that all the south-western Saharan *Acomys* considered in the present study belong to the same species.

According to Denys *et al.* (1994), *A. airensis* can be separated from *A. chudeaui* based on its wider inter-orbital constriction, the absence of an antero-central cusp (tmA) on the prelobe of lower M1, a narrower braincase and larger nasals. However, the morphometrical differences observed by these authors between species should be considered with caution because only an unknown number of specimens of *A. airensis* was compared with specimens of *A. chudeaui*, and thus did not take into account intra-specific variability. Our sample sizes for specimens of *A. chudeaui* are too small to rigorously compare the cranial measurements of these two species. Moreover, Denys *et al.* (1994: 221) state that *A. airensis* has no tmA on the lower M1, but a tmA is present on two of the four lower M1 drawn in their Fig. 4 (Denys *et al.*, 1994: 223). A tmA on the lower M1 was observed not only in most specimens of clade IV (i.e. *A. chudeaui*, 13 of 15), but also in most specimens of other clades (31 of 43), suggesting that this character is not suitable for discriminating between the two species. A tmA was also observed on the lower M1 in two of the three *A.*

chudeaui from Tata (Morocco) housed in the Muséum National d'Histoire Naturelle (MNHN) collections (V. Nicolas, pers. observ.).

According to Denys *et al.* (1994), these species can also be differentiated on their cusp pattern. The cusps are well-differentiated from each others (bunodont pattern) in *A. airensis*, whereas they are less individualized, more or less fused, with no longitudinal crests in *A. chudeaui*. However, although Denys *et al.* (1994: 218) state that '*A. airensis* can be distinguished by a rather bunodont pattern of cusps which are well separated from each other, especially the t1, t4 and t9 on the upper M1', they also reveal in the same paragraph that 't1 and t4 are fused to t2 and t5'. Consequently, such a cusp pattern does not really appear either as a valuable character to differentiate *A. airensis* from *A. chudeaui*. Finally, according to Denys *et al.* (1994), the t8 and t9 of upper M1 are well-fused at early stages of wear in *A. chudeaui*. However we observed on the specimens sequenced that the t8 and t9 of upper M1 are well separated at early stages of wear not only in clade IV (i.e. *A. chudeaui*), but also in clades II (*A. airensis*) and III. The same was true for all specimens collected in Moudjéria (Mauritania) and Tata (Morocco) and identified in the MNHN collections as *A. chudeaui* (V. Nicolas, pers. observ.). Thus, this character should neither be considered as diagnostic between the two taxa because it appears to be more related to the degree of tooth wear than to a valid species specific character.

As a conclusion, the dental characters proposed by Denys *et al.* (1994) to distinguish the two species can no longer be considered as valid. After a careful comparison of the skulls of all our sequenced specimens, no character was found to discriminate the specimens of clade IV (representing topotypical materials of *A. chudeaui*) from specimens of the other clades (V. Nicolas, pers. observ.). Moreover, the observation of three specimens from Tata (Morocco) suggests that the same species is also present in this region. However this would have to be confirmed by molecular data and/or morphometrical analyses based on a larger sample size.

To conclude, cytogenetical, molecular, and morphological analyses all concur in suggesting that *A. airensis* should be considered as a junior synonym of *A. chudeaui*, the distribution of which would range from Morocco and Mauritania to at least eastern Niger across Mali, and probably eastwards to Chad *sensu* Musser & Carleton (2005).

PHYLOGEOGRAPHY

The ML, Bayesian and network analyses clearly showed a strong genetic structure within southwestern Saharan *Acomys*, with several phylogroups

separated by moderate to high genetic distances (Table 2). Moreover, these phylogroups display strictly non-overlapping geographic distributions (Fig. 1) and, in general, specimens from a given clade originate from adjacent geographical localities. However, the observed pattern is not simply the result of a phenomenon of isolation-by-distance: although genetic differentiation is generally high between geographically distant populations (as expected for such a small rodent species, with low dispersal ability), differentiation is either low or high between neighbouring populations. The presence of spatially circumscribed haplogroups separated by relatively large mutational distances corresponds to the category 1 phylogeographic pattern observed for small- or medium-sized terrestrial mammals on almost all continents (Avise, 2000). A likely explanation for these major phylogeographic discontinuities is the presence of long-term extrinsic barriers to genetic exchange between regions. After a sufficiently long time, allopatric populations will occupy recognizable, deeply-separated branches in an intraspecific gene tree. The amount of genetic differentiation will reflect the mutations accumulated after the separation of these populations, and/or effects of lineage sorting from a polymorphic ancestral gene pool.

To identify potential extrinsic barriers to gene flow, it is necessary to have a good knowledge of the habitat preferences of the species. *Acomys* has often been considered to be strictly confined to rocky habitats, resulting in a patchy distribution with many populations isolated on cliffs and outcrops (Rosevear, 1969; Sicard & Tranier, 1996). However, several studies revealed that it could also be found in gardens, grain storage units, straw huts, and even sandy areas (Tranier *et al.*, 1999; Bâ *et al.*, 2001; Dobigny *et al.*, 2001b). Large sandy areas do appear to be unfavourable to *Acomys* anyway, and this could explain the genetic distance observed between clades IV and III and between clades II and I (Fig. 1). Specimens from Fachi (clade I) were found commensal, in an oasis closed to a rocky area (several hundreds of meters to more than 1 km), within the Tenere sand dunes field and isolated by at least 100–150 km from the nearest rocky areas. This population could have an anthropogenic origin or, alternatively, could be the relic of a wider distribution. Owing to the large genetic distances between the specimens from Fachi and all other specimens, and to the TMRCA estimated at 62 000 years (range 5000–118 000) for these specimens, an anthropogenic origin for this population is extremely doubtful.

As for the geographical limit between clades II and III, its northern part matches the relief of the region: There, *Acomys* appears to be restricted to areas of more than 500 m a.s.l., and is absent from lowland

areas. However, the distinction between these two clades is only moderately supported by the *cytb* sequence data, whereas the D-loop sequence data suggest that clade II is derived from within clade III. The genus *Acomys* is known from the mountains of southern Algeria (Kowalski & Rzebik-Kowalska, 1991) as *A. seurati* (Musser & Carleton, 2005) and, to better understand the phylogeographic structure observed between the Adrar des Iforas and the Air massifs, it would be necessary to add samples from these populations from southern Algeria. Moreover, although *Acomys* is restricted to areas above 500 m a.s.l. in the northern parts of Mali and Niger, this is not true in the south (Fig. 1).

To identify potential extrinsic barriers to gene flow, it is often useful to compare the phylogeographic patterns obtained for several species with a similar geographical distribution. Comparison of our results with those obtained for other taxa is difficult because only few phylogeographic studies cover a geographical range similar to that employed in the present study. In *Mastomys erythroleucus* Brouat *et al.* (2009) found two distinct phylogroups in south-western Niger and central Niger, respectively, the limit of which corresponds exactly to the southern limit between our *Acomys* clades II and III. Moreover, Dobigny *et al.* (2005) showed that these two geographical areas are characterized by the presence of distinct species of the genus *Taterillus*. It is striking to find this common pattern of geographical separation in species with very different ecological requirements. *Mastomys erythroleucus* is found in a variety of habitats in the Sahelo–Sudanian bioclimatic area, including grasslands, fields, and near or inside human habitations (Duplantier & Granjon, 1988), whereas *Taterillus* species range from subdesert to Sahelo–Sudanian savannas (Nowak, 1999). These independently obtained pieces of evidence strongly suggest that past environmental events are responsible for the observed pattern of genetic divergence because no current geographical feature can explain it.

There is undoubtedly a wide error attached with any divergence date estimates based on DNA sequence data, and ‘divergence time estimates should be considered as temporal hypotheses amenable to future empirical testing’ (Martinez-Solano *et al.*, 2006); however, when used wisely, they can provide insight into phylogenetic interpretations (Hewitt, 1996). Coalescence within all of the major clades of south-western Saharan *Acomys* clearly occurred during Middle Pleistocene and/or the beginning of Late Pleistocene. This result corroborates those obtained for other West African rodent species (Mouline *et al.*, 2008; Nicolas *et al.*, 2008a) and is consistent with the meta-analyses of Avise & Walker (1998) that illustrated the role of Pleistocene condi-

tions in the phylogeographic patterns of Holarctic vertebrate species (Avise & Walker, 1998). During the Pleistocene, the Sahara has experienced drastic climatic changes, rendering its boundaries highly variable (Rognon, 1989). It could at least partly explain the geographical structure of *Acomys* mitochondrial clades: populations of *Acomys* could have been restricted to separate refugia during phases of unfavourable periods of the Pleistocene, and have experienced subsequent expansion during phases of favourable habitat expansion. In agreement with this hypothesis, a signal of population expansion was recorded in clade III, and possibly also in clade II. Additionally, interspecific competition with other rodents, such as arid-adapted gerbils, could have reinforced the exclusion of *Acomys* from sandy areas during arid periods; by contrast, competition between *Acomys* and these gerbilline rodents was probably limited during more humid periods unsuitable for these sand-dweller species. At present, however, it is possible that the development of human settlements and their associated resources facilitates the wide distribution of *Acomys* even outside rocky massifs. Increased sample size as well as more precise data about ecological requirements and population connectivity in spiny mice will be required for further testing of these aspects of their phylogeographic history.

CONCLUSIONS

Based on a comprehensive sampling and the use of an integrative approach (combining cytogenetic, molecular and morphological data), we firmly propose that *A. airensis* should be considered as a junior synonym of *A. chudeaui*. Distribution would range from Morocco and Mauritania to at least eastern Niger across Mali, and probably eastwards to Chad *sensu* Musser & Carleton (2005).

In this species, mostly found in rocky habitats but able to survive in nonrocky habitats (probably depending on food resource availability), a phenomenon of isolation-by-distance was observed within our entire data set, suggesting that this species has low dispersal abilities. However, its wide geographical distribution shows that it has been able to colonize all the rocky areas of south-western Sahara, even if they are not connected to one another. The correlation between geographical and genetic distances was not significant within clades II and III, suggesting that *Acomys* has been able to disperse at the geological scale to some extent. All these results suggest that, during arid or semi-arid paleoclimatic periods, when large sandy areas were present, *Acomys* was restricted to rocky massifs. In more humid periods, however, when savannah and/or steppe habitat pre-

vailed, this species was able to disperse and to survive outside rocky areas because food resources were abundant, and competition with arid-adapted species from sandy habitats was potentially less. Additional sampling (especially in southern Algeria) and increased sample sizes in each locality, as well as a better knowledge of the ecological requirements of this species, are necessary to test these hypotheses, for a better understanding of its phylogeographic history.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Table S1. List of specimens used in the present study with geographical origin, haplotypes and Genbank numbers, and cytogenetical data.

Figure S2. Plots of pairwise numbers of observed differences against the corresponding values for inferred substitutions. Black circles, transitions; white circles, transversions. All the ingroup sequences and the five outgroups retained in the final phylogenetic analyses were included.

Figure S3. Phylogeny recovered by the maximum likelihood (ML) analysis for the cytochrome *b* gene and D-loop region (GTR + I + G) combined. Numbers above nodes represent ML bootstrap support (500 replications) and Bayesian posterior probabilities respectively. To improve clarity, posterior probabilities for the most apical nodes are not shown.

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